

Standard Operating Procedure (SOP)

Application of Graphene Coating on Anode and Cathode of a 200 mAh Li-ion Cell

Purpose

To prepare and coat graphene-enhanced anode and cathode electrodes for a 200 mAh lithium-ion coin/pouch cell to enhance conductivity, energy density, and cycle stability.

1. Materials Required

For Anode (typically Graphite-based):

Graphite powder
Conductive graphene nanoplatelets or reduced graphene oxide (rGO)
PVDF binder (polyvinylidene fluoride)
N-Methyl-2-pyrrolidone (NMP) solvent
Copper foil (current collector)

For Cathode (typically LFP/NMC-based):

LiFePO₄ or NMC powder
Conductive graphene or reduced graphene oxide
Carbon black (e.g., Super P)
PVDF binder
NMP solvent
Aluminum foil (current collector)

2. Equipment Required

Magnetic stirrer or planetary mixer
Ultrasonicator
Doctor blade coater
Vacuum oven
Glovebox (Ar or N₂ atmosphere)
Precision balance
Roll press or calendaring machine
Punching machine (for electrode discs or strips)

3. Procedure

Step 1: Preparation of Anode Slurry

Weigh materials (example ratio by weight: Graphite 85%, Graphene 5%, PVDF 10%).
Disperse graphene in NMP using ultrasonicator for 30–60 minutes.
Add graphite and stir for 2 hours until homogeneous.

Dissolve PVDF in NMP separately, then mix with the active material.
Continue mixing for 12 hours at room temperature.

Step 2: Coating Anode

Coat slurry onto copper foil using a doctor blade (wet thickness $\sim 100\text{ }\mu\text{m}$).
Dry in vacuum oven at 80°C for 12 hours.
Calender the electrode to desired thickness (e.g., $\sim 50\text{--}70\text{ }\mu\text{m}$).
Punch electrode discs (typically 12–14 mm diameter for 200 mAh cells).
Dry again at 120°C under vacuum for 4 hours before transferring to glovebox.

Step 3: Preparation of Cathode Slurry

Weigh materials (e.g., LFP/NMC 80%, Graphene $\leq 5\%$, Carbon black $\leq 5\%$, PVDF 10%).
Disperse graphene and carbon black in NMP via ultrasonication.
Add LFP/NMC powder and stir well for uniform dispersion.
Add PVDF binder, continue mixing for 12 hours.

Step 4: Coating Cathode

Coat slurry on aluminum foil using a doctor blade.
Dry in vacuum oven at 80°C for 12 hours.
Calender the electrode to uniform thickness ($\sim 50\text{--}80\text{ }\mu\text{m}$).
Punch cathode discs, similar to anode.
Final drying at 120°C under vacuum before cell assembly.

4. Cell Assembly

Transfer electrodes to glovebox (H_2O , $\text{O}_2 < 1\text{ ppm}$).
Assemble the 200 mAh cell in coin cell or pouch cell format using:

- Graphene-coated anode
- Separator (e.g., Celgard or glass fiber)
- Electrolyte (e.g., 1M LiPF_6 in EC/DMC)
- Graphene-coated cathode

Seal cell and allow for electrolyte soaking (12 hrs resting).

5. Formation & Testing

The assembled cells should undergo initial formation through charge-discharge cycling at 0.1C for 3 to 5 cycles to stabilize the solid electrolyte interface (SEI) and ensure consistent electrochemical behavior.

Subsequently, performance evaluation should include capacity measurement, rate capability testing, electrochemical impedance spectroscopy (EIS), and initial cycle life assessment.

After baseline performance is established, the cells can be tested for fast-charging capability at rates up to 3C–5C to evaluate thermal and electrochemical stability under high current loads.

Long-term cycle life testing should be conducted for up to 500 cycles under standard operating conditions. Additionally, accelerated simulation testing can be performed to project durability over 5,000 cycles for lifetime estimation and reliability assessment.

6. Safety Precautions

Handle NMP and lithium salts with gloves and goggles.

Use fume hood for solvent handling.

Perform assembly inside glovebox only.

Dispose of graphene and nanomaterials safely as per lab protocols.